



UTM
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Faculty of Computing

Project Task 3

(Choosing The Appropriate LAN Devices)

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Prepared by : 1) Yong Wern Jie (A22EC0121)
2) Tan Jun Yuan (A22EC0107)
3) Chen Pyng Haw (A22EC0042)
4) Koh Li Hui (A22EC0059)

Section : 02

Group : 3 – NotAvailable

Lecturer : Dr. Muhalim Bin Mohamed Amin

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3.1 Introduction to The Needed Devices

1. Router

A router is a device that connects computer networks and forwards data packets between them. It acts as a traffic director, determining the best path for data to travel from one network to another. When data is sent through a network, it is divided into smaller packets. These packets are then forwarded from one router to another until they reach their destination. Routers use routing tables or policies to determine the next network for the packet to be sent to. They are connected to multiple data lines from different IP networks, allowing them to direct packets to the appropriate destination. Routers play a crucial role in the functioning of the internet and are essential for communication between networks.

2. Switch

A network switch is a small and important device that centralizes communications among several connected devices such as computers or other network devices in one local area network (LAN) which enables them to talk by exchanging data packets. Network devices are made with the help of switches that carry a large amount of traffic in telecommunications. Switches can be hardware devices that manage physical networks or software-based virtual devices. Besides that, switches have many significant ports, and when data from the network devices arrives at any port, the destination address is examined first, and some checks are also done and then it is processed back to the devices. In the Open Systems Interconnection (OSI) model, a network switch operates on the data-link layer, or Layer 2.

3. Patch Panel

A patch panel in a local area network (LAN) is a mounted hardware assembly that contains ports that are used to connect and manage incoming and outgoing LAN cables. A patch panel provides a way to keep large numbers of cables organized, enabling flexible connectivity into network hardware located in a data center or an access or wiring closet.

4. Cable

Cable is a fundamental component in computer networks, serving as the crucial link between networks and end-users as well as used to link routers, switches, and servers. Even in wireless technologies, cables play a vital role in backbone connections to access points. The evolution of cable technology, from twisted pair cables in telegraphs to coaxial cables and optical fibers in video services, has been driven by the internet's influence, necessitating networks to support multi-service applications such as voice, video, and data.

5. Wireless Access Point

A wireless access point (AP) serves as a pivotal device facilitating wireless connectivity by enabling wireless devices to access a wired network and connect to the internet without physical connections. Operating in accordance with the IEEE 802.11 standard, APs act as bridges, linking wireless devices to the network infrastructure, and are integral components in open WiFi networks, delivering mobile internet access. In the specific context of research at Wrocław University of Technology in Poland, the PWR-WiFi network relies on APs to provide essential connectivity for students, employees, and other users. The research paper focuses on analyzing the performance of 12 indoor APs within the university's wireless network, located in a five-storey building on the main campus, with the aim of understanding their loads and efficiency in enhancing network functionality.

6. Modem

A modem transforms digital information from your computer into analog signals that can transmit over wires (and vice versa) by modulating and demodulating electrical impulses sent through phone lines, coaxial cables, or other types of wiring. Most standalone modems have just two ports: one that connects to the outside world, and an Ethernet jack that connects to a computer or a router.

7. Server

A server is computer hardware or software that provides functionality for other programs or devices, known as "clients". It is a process that serves data for clients in a client-server

model, where communication is based on request and response. Servers can provide various services such as sharing data or resources among multiple clients or performing computations for a client. They can be specialized for running specific types of servers, such as database servers, file servers, mail servers, web servers, and application servers.

8. Firewall

A hardware firewall is one of the most efficient network security solutions to protect network devices against attacks designed to exploit the underlying operating system or the programs running alongside it. It acts as a physical appliance which is deployed to enforce a network boundary. All network links crossing this boundary pass through this firewall, which enables it to perform inspection of both inbound and outbound network traffic and enforce access controls and other security policies. Firewalls commonly are configured to block traffic over any unused or undesirable ports from entering the network and to stop certain types of traffic from leaving the network such as traffic that could leak sensitive data.

9. Personal Computer (PC)

Personal computer (PC) is a multi-purpose microcomputer designed for individual use, with a size, capabilities, and price that make it feasible for personal use. PCs are intended to be operated directly by end users, without the need for computer experts or technicians. Personal computers typically consist of a computer case, a power supply, a motherboard, storage devices (such as hard disk drives or solid-state drives), and often an optical disc drive. External devices such as monitors, keyboards, and mice are commonly used with personal computers.

10. Monitor

A monitor is an output device that displays information in pictorial or textual form. It typically consists of a visual display, support electronics, power supply, housing, electrical connectors, and external user controls. In modern monitors, the display is usually an LCD with LED backlight, replacing the older CRT technology used before the mid-2000s.

Originally, computer monitors were used for data processing while televisions were used for video. However, from the 1980s onward, computers and their monitors have been used for both data processing and video, and televisions have incorporated computer functionality as well. As a result, modern computer monitors and televisions are often functionally interchangeable. The size of a monitor is typically measured diagonally, and the actual viewable image size may vary depending on the aspect ratio and the design of the unit.

11. Projector

A projector is an output device that takes images generated by a computer or Blu-ray player and reproduces them by projection onto a screen, wall, or another surface. In most cases, the surface projected onto is large, flat, and lightly colored. For example, you could use a projector to show a presentation on a large screen so that everyone in the room can see it. Projectors can produce either still (slides) or moving images (videos). A projector is often about the size of a toaster and weighs only a few pounds.

12. Smart TV

A Smart TV is a television equipped with integrated Internet capabilities and the ability to run dedicated software applications. It functions as a convergence of a traditional TV terminal, an operating system, and digital content, all seamlessly connected to the Internet. These TVs offer diverse digital services encompassing multimedia, gaming, internet browsing, on-demand entertainment, and online interactive sessions. As technology advances, Smart TVs are anticipated to evolve into more intelligent, interactive, and versatile platforms. The testing of smart TV applications differs from that of mobile apps, primarily due to the distinct nature of user interaction and the development environment. Interaction with smart TV apps is typically through a remote controller, shaping the testing methodologies. The escalating demand for smart TV applications, particularly in the context of the growing prevalence of Internet of Things (IoT) services, underscores the imperative to develop high-quality applications for these innovative television systems.

13. Printer

Printer is an electronic device that produces a hard copy or printed version of electronic documents, images, or other digital content. Printers are commonly used to transfer information from a digital format to a tangible, paper-based format. There are a few different ways a printer can connect to and communicate with a computer. It can be connected directly to the computer or indirectly via a network. A printer may be Ethernet enabled and be cabled to an Ethernet switch, or it may connect to a Wi-Fi (wireless) network, or both. Till today, the most common connection types are by USB (universal serial bus), cable (wired) or via Wi-Fi (wireless). It is widely used for a range of applications, from producing documents for personal use to handling high-volume printing in professional environments including homes, offices, educational institutions, and businesses.

14. Connector

A connector is a device that terminates a segment of cabling or provides an entry point for network devices such as computers, hubs and routers. A registered jack (RJ) is a standardized physical network interface for connecting telecommunications or data equipment. The physical connectors that registered jacks use are mainly of the modular connector and 50-pin miniature ribbon connector types. The most common twisted-pair connector is an 8-position, 8-contact (8P8C) modular plug and jack commonly referred to as an RJ45 connector.

15. Fiber Optic Cable

Fiber optic cable is an assembly similar to an electrical cable but contains one or more optical fibers that are used to carry light. The cable consists of a core and a cladding layer, with the cladding usually coated with a layer of acrylate polymer or polyimide. This coating protects the fiber from damage but does not contribute to its optical waveguide properties. The individual coated fibers are then surrounded by a tough resin buffer layer or core tube(s) to form the cable core. Multiple layers of protective sheathing are added to the cable depending on the application. Fiber optic cables are used for various optical

communication applications, such as long-distance telecommunication or high-speed data connections within a building.

3.2 Table of Devices

Devices	Description	Price per unit (RM)	Quantity	Total price (RM)
Router  Figure 3.2.1 Router	<u>TP-Link Archer AX73</u> - Dual-band Wi-Fi 6 router: Supports the latest Wi-Fi 6 technology for faster speeds and better performance. - Speeds of up to 5400 Mbps: Provides enough bandwidth for even the most demanding applications. - Connects over 200 devices: Can handle many connected devices without sacrificing performance. - HomeShield security: Protects your network from malware and other threats. - 1.5 GHz Triple-Core CPU, 512 MB of RAM, 1x Gigabit WAN port, and 4x Gigabit LAN ports.	549	5	2745
Switch  Figure 3.2.2 Switch	<u>S3400-24T4SP</u> - Improved management: Virtual deployment of dual switches offers a simple, flexible, and high-performance network for serving all scenarios on campus. - Sound Security Protection Policies: Support to provide equipment-level security and level-based command line protection. - Easy Network Maintenance: Support Web management, Support firmware upgrade and configuration file upload/download via TFTP/HTTP/FTP server, Support firmware auto upgrade Remote Monitoring (RMON). - 24× PoE+ 10/100/1000 Mbps RJ45	2033	8	16264

	ports provide up to 30W per port and total PoE power budget of 370W, together with 4× Gigabit SFP slots for high-speed optical connections. - 4 years limited warranty.			
Patch Panel  Figure 3.2.3 Patch Panel	<u>Rapink Patch Panel 24 Port Cat6 with Inline Keystone 10G Support</u> - Toolless termination: This Cat6 patch panel has more expansion and upgrade capabilities than 12 port patch panel and 16 port patch panel, and can use two of 24 port patch panel cat6 instead of 48 port patch panel. - Easy expansion and repair: Its patch panel holder allows easy input and output cat6 coupler. Therefore, it is convenient to repair when 1 port network is broken. - 10G speed via Gold plated pins: All 8P8C of rj45 jack panel keystone are golden plated to support better network connection up to 10G speed Ethernet network. - Detachable back bar included: With zip ties, it can effectively reduce the deformation of the ethernet cable, improve the working stability of the network connection port, and avoid keystone loose and the occurrence of network failures caused by the loosening and deformation of the cable	199.85	8	1598.8
Cat 6 Ethernet Cable  Figure 3.2.4 Cable	<u>Cable Matters 10Gbps Snagless Long Cat 6 Ethernet Cable</u> - Brand: Cable Matters - Connector Type: RJ45 - Cable Type: Ethernet	142.70/ 1524 cm	197	28111.90

	- Compatible Devices: Server, Router, Personal Computer, Printer - Compatible Phone Models: VoIP phones - Data transfer speed :10 Gigabit - Rating: 550Mhz - Specification: Gold-plated contact, snagless design, UL Certified, Molded strain relief, flexible PVC jacket, stranded twisted pair, pair separator, bare copper conductor - Application/Compatibility: Classroom, meeting room, server room, office, printer, laptop, computer, router, tv			
Modem  Figure 3.2.4 Modem	<u>ARRIS SURFboard Gigabit DOCSIS 3.1 Cable Modem</u> - DOCSIS 3.1 cable modem, backward compatible with DOCSIS 3.0. Supports IPv4 and IPv6 Internet browsing standards. - As a Docsis 3.1 32X8 Modem capable of up to 10 Gbps download speeds. As a Docsis 3.0 32x8 Modem is capable of up to 1.4 Gbps download speeds. - Gigabit Ethernet ports. Capable of supporting 2 Home Networks, both capable of gigabit speeds.	1007.23	2	2014.46

Wireless Access Point  Figure 3.2.6 Wireless Access Point	<u>TP-Link EAP670</u> - Special Feature: Access Point Mode, WPS - Frequency Band Class: Dual-Band - Wireless Communication Standard: 802.11ax - Compatible Devices: Personal Computer, phone device, laptop - Frequency: 5 GHz - Recommended Uses for Product: Large building - Included Components: Installation Guide, Power Adapter, Ceiling/Wall Mounting Kits, EAP670 - Connectivity Technology: Ethernet - Speed: 5400 Mbps - Coverage area: 204 m²	657.54	5	3287.70
Server  Figure 3.2.7 Server	<u>Cisco UCS C480M5 High-Performance Rack Server</u> - Form factor: 4U rackmount - Processors: Dual Intel Xeon Scalable processors (2nd generation) - Memory: Up to 24 x 16GB DDR4 DIMMs for a total of 384GB - Storage: Up to 32x 2.5-inch hot-swappable SAS/SATA HDDs or SSDs - Networking: Dual 10GBASE-T Intel x550 network adapters - High-performance PCIe Gen 3 slots	20000	5	100000

	- Support for up to 12 NVMe SSDs - Integrated management controller (iMC) - Cisco UCS Manager software			
Firewall  Figure 3.2.8 Firewall	<u>FS NSG-5100 Next-Generation Firewall</u> - Up to 6 Gbps of Firewall Performance. - Comprehensive Threat Detection and Prevention: Provide real-time protection for applications from network attacks including viruses, spyware, worms, botnets, ARP spoofing, DoS/DDoS, Trojans, buffer overflows, and SQL injections. - Granular Application Identification and Control: Identify and prevent potential threats associated with high-risk applications while providing policy-based control over applications, users, and user-groups. - 6x GE (RJ45), 4x SFP ports, 1 AC power unit, IPS service, Anti-Virus, QoS, URL filtering service, hardware warranty and software upgrading service for 2 years. - 2 years limited warranty.	36811	5	184055
Personal Computer (PC)  Figure 3.2.9 PC	<u>HP EliteDesk 800 G9 Desktop</u> - Processor: Up to 13th Gen Intel® Core™ i9 processors - Memory: Up to 64GB DDR5 memory - Storage: Up to 8TB NVMe SSD - Operating System: Windows 11 Pro, Windows 10 Pro - Form Factor: Tower, Small Form Factor	3699	156	577044

	- Security: HP Wolf Security for Business - Manageability: HP Intelligent Management Suite			
Monitor  Figure 3.2.10 Monitor	<u>HP V24i G5 FHD Monitor</u> - Display size: 23.8" - Display type: IPS - Panel active area (inches): 20.75 x 11.67 in - Panel active area (cms): 527.04 x 296.46 mm - Brightness: 250 nits - Pixel Pitch: 0.274 mm - Input connector: 1 VGA; 1 HDMI 1.4 (with HDCP support) - Scan range (horizontal): 30-80 KHz - Scan range (vertical): 50-60 Hz - Recommended resolution: 1920 x 1080 (FHD) - Aspect ratio: 16:9 - Static contrast ratio; Dynamic contrast ratio: 1000:1 static - Response time: 5ms GtG - Power consumption: 26 W (maximum), 23.5 W (typical), 0.5 W (standby) - Power supply: Input voltage 100 to 240 VAC - Dimensions: 53.96 x 4.76 x 32.5 cm - Weight: 3.46 kg	750	156	117000

Projector  Figure 3.2.11 Projector	<u>Epson VS250 SVGA</u> - Special Feature: Portable, Lightweight - Connectivity Technology: HDMI - Display resolution: SVGA - SVGA (800 x 600) (native) / 1400 x 1050 (resized) - Maximum Display Resolution: 800 x 600 - Brightness: 3,200 lumens for accurate and vivid color even in well-lit rooms - SVGA resolution (800 x 600) for everyday presentations and graphics - Projected distance: 30 inches to 350 inches (0.88 meters to 10.44 meters)	2506.21	8	20049.68
Smart TV  Figure 3.2.12 Smart TV	<u>Samsung-3P QN900B</u> - Screen size: 75 Inches - Supported Internet services: Netflix, Amazon Instant Video, YouTube - Display technology: QLED - Product dimensions: 31.8D x 165.5W x 101.7H centimeters - Resolution: 7,680 x 4,320 - Refresh rate: 120 - Special features: Flat - Quantum Matrix Technology Pro - Dolby Atmos experience with top channel speakers - Connectivity: HDMI x 4, USB x3, HDMI (High Frame Rate) 4K 120Hz (for HDMI 1/2/3/4), Ethernet (LAN), Digital Audio Out (Optical), eARC (HDMI3)	44,126	1	44,126

	- Sound Output (RMS): 90W - VESA Spec: 400 x 400 mm - Set Size without Stand (WxHxD): 1654.8 x 945.6 x 15.4 mm - Set Size with Stand (WxHxD): 1654.8 x 1017.4 x 318.3 mm			
Printer  Figure 3.2.13 Printer	<u>Color imageCLASS MF751Cdw</u> - Categories: Multifunction, Wireless, Color Laser Printer - Functions: Print, Scan, Copy - Document Feeder: 50 Sheet ADF, Simplex Scan - Print/Copy Resolution: Up to 1200 x 1200 dots per inch (dpi) - Scan Resolution: Up to 600 x 600 dots per inch (dpi) - Network Security: IP/MAC Address Filtering, IPSEC, TLS 1.3, SNMP V3.0, IEEE 802.1X, IPv4/IPv6, SMTP Authentication, POP3 - Connectivity: USB 2.0 High-Speed, 10/100/1000Base-T/TX Ethernet (Network), Wi-Fi 802.11b/g/n, Wi-Fi® Direct Connection - USB Direct Print: JPG, TIFF, PDF - Operating Systems: Windows® 11 / 10 / 8.1 / Windows® Server 2012 / 2012 R2 / 2016 / 2019 / 2022, Mac OS X (10.11 or later) - Unit Dimensions: 16.8" (W) x 18.2" (D) x 17" (H) - Unit Weight: 22kg	1874	2	3748

Connector Cost-Effective 200-Pack 200 Pack  Figure 3.2.14 Monitor	<u>Cable Matters 200-Pack Cat 6 Pass Through RJ45 Connectors</u> - Brand: Cable Matters - Connector Type: RJ45 - Cat 6 RJ45 pass through connector for custom-length cables - Supports 23-26 AWG round or flat stranded wire	129.97/ pack 1 pack=200	2	255.94
Fiber Optic Cable  SC UPC Single Mode 3m	<u>SC-UPC APC Pigtail</u> - Fiber Type: Single Mode - Fiber Mode: SMF 9/125um - Insertion Loss: 0.15dB - Return Loss: 53.6dB - Fiber Class:G.657A - Compliant: RoSH, ISO9001 - Wavelength: 1310/1550nm	1.21/1m	300	363
Total				1100663 .48

3.3 Reflections

3.3.1 Are you surprised by the prices? How were you surprised?

We are astounded by the price of RM 1100663.48, which is half of our overall budget. This unexpected expense caught us off guard, as we never anticipated that acquiring these network devices and equipment would be so costly, especially when most of the options are not even the top or second-best choices. Even seemingly inexpensive items like cables surprised us with a total cost of RM 28111.90. We were surprised by the overall cost of these items, particularly given that the additional costs of items like keyboards and mice were not even discussed.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

Cost was not our primary consideration when selecting networking devices. Instead, we prioritized finding devices that were best suited for our project's requirements and would optimize network efficiency within our budget. We carefully evaluated various networking device options and selected those that offered the most suitable balance of performance, cost-effectiveness, and alignment with our project goals.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Router:

For the routers used in this project, we have chosen [TP-Link Archer AX73](#). The major difference between TP-Link Archer AX73 with other different brands we can compare is NETGEAR R6700AX.

Specification	NETGEAR R6700AX	TP-Link Archer AX73
Maximum Wi-Fi Speed	1,800 Mbps	5,400 Mbps
USB Ports	1x USB 2.0	1x USB 3.0 Type-A + 1x USB 2.0 Type-A

Based on the table above, the TP-Link Archer AX73 is a superior router to the NETGEAR R6700AX in all but one way. It boasts faster maximum Wi-Fi speed, resulting in faster downloads and uploads, smoother streaming, and a better online gaming experience. Additionally, its broader range ensures wider coverage and better signal penetration. Moreover, it offers more LAN ports for connecting more wired devices and a USB 3.0 Type-A port that allows you to share files and create a NAS device. Finally, the TP-Link Archer AX73 is available at a more affordable price.

Switch:

For the switches used in this project, we have chosen FS S3400-24T4FP. The major difference between FS S3400-24T4FP with other different brands we can compare is NETGEAR JGS524.

Specification	FS S3400-24T4FP	NETGEAR JGS524
Packet Buffer Size	500KB	256KB
Cooling	3 Built-in Fans	Fanless
Power Consumption	400W	17.9W
Ability to Handle Traffic	High	Limited
Cooling Functionality	Yes	No
Power Efficiency	High	Low

Based on the comparison table above, FS S3400-24T4FP has a higher size of the packet buffer. The bigger the size of the packet buffer, the higher the ability to handle different traffic patterns and avoid packet loss during periods of congestion. Besides that, FS S3400-24T4FP has 3 built-in fans inside it. This indicates that FS S3400-24T4FP offers cooling functionality that can help to prevent overheating during operation and ensures the reliable functioning of the switch. In addition to that, the power consumption of the FS S3400-24T4FP switch, 400W, is higher than NETGEAR JGS524 switch. Thus, the FS S3400-24T4FP network switch will become the most

suitable selection for us as it can give higher performance among many workstations in the faculty building.

Patch Panel:

We have chosen [Rapink Patch Panel 24 Port Cat6 with Inline Keystone](#) for the patch panels of this project. The major difference between Rapink Patch Panel and other different brands we can compare is TRENDnet 24-Port Blank Keystone 1U Patch Panel.

Specification	Rapink Patch Panel	TRENDnet Patch Panel
Connector Types	VGA, HDMI, RJ45	RJ45
Cable Types	HDMI, VGA, Ethernet (Cat 6, Cat5e, Cat7)	Ethernet (Cat 6)
Cable Management	Detachable back bar with zip ties	Basic
10G Support	Yes	No
Dimensions	20 x 4.5 x 2.5 inches	19 x 1.77 x 0.4 inches
Price	RM 199.85	RM 83.71

Based on the table above, the Rapink Patch Panel offers a broader range of features, including support for multiple cable types, better cable management, 10G support, and a wider range of connector types. While it comes at a higher price point compared to the TRENDnet panel, the additional features make it a more versatile and robust choice for most network setups. Therefore, we chose Rapink Patch Panel as our patch panel.

Cable:

Cable Matters 10Gbps Snagless Long Cat 6 Ethernet Cable emerges as the preferred choice for my project due to several key reasons when compared to Cat-6 Ethernet Patch Cable from Amazon Basics. Firstly, Cable Matters 10Gbps Snagless Long Cat 6 Ethernet Cable is Cat6 rated and constructed with 24 AWG copper wire, ensuring higher performance and bandwidth compared to

Cat-6 Ethernet Patch Cable that use normal features. Besides, Cable Matters 10Gbps Snagless Long Cat 6 Ethernet Cable Can support 10-Gigabit Ethernet while Cat-6 Ethernet Patch Cable only supports 1-Gigabit Ethernet. Additionally, Cable 2 is explicitly designed for a diverse range of network components, including servers, routers, personal computers, printers, and VoIP phones, making it versatile for various project requirements while Cat-6 Ethernet Patch Cable can only be compatible with less devices. Overall, the comprehensive performance features and adaptability of Cable 2 make it a superior choice for our project needs.

Modem:

We selected [ARRIS SURFboard Gigabit DOCSIS 3.1 Cable Modem](#) for the modem in this project. The major difference between ARRIS SURFboard Gigabit DOCSIS 3.1 Cable Modem and other different brands we can compare is Motorola MB8600 DOCSIS 3.1 Cable Modem

Specification	ARRIS SURFboard SB8200	Motorola MB8600
DOCSIS Standard	DOCSIS 3.1, also compatible with DOCSIS 3.0	DOCSIS 3.1, DOCSIS 3.0 32×8
Top Internet Speed Supported	Up to 2 Gbps*	Up to 6 Gbps
Ethernet Ports	Two 1-Gigabit Ethernet ports	1x 1Gbps LAN
Compatibility with Providers	Compatible with major U.S. cable providers	Comcast Xfinity, Cox, and others
Additional Features	Ideal for 4K Ultra-HD video streaming and VR gaming, Energy-Efficient Ethernet	WAN Link Aggregation, Multi-Static-IP support

Based on the table above, the ARRIS SURFboard SB8200 is a preferred choice due to its dual 1-Gigabit Ethernet ports, offering the flexibility of setting up two networks or a failover connection. It is highly compatible with major U.S. cable providers, ensuring broad usability. The modem excels in handling high-demand applications like 4K Ultra-HD video streaming and VR gaming, catering to users with intensive internet usage. Additionally, its energy-efficient ethernet feature presents potential cost savings. The inclusion of an app for easy modem setup further enhances its user-friendliness, making it an attractive option for those seeking a blend of performance, versatility, and convenience.

Access point:

The [TP-Link EAP670](#) stands out as the preferred choice due to its advanced features and superior performance when compared to the NETGEAR WAC104. Supporting the latest Wi-Fi 6 standard, the EAP670 offers blazing-fast dual-band speeds of up to 5400 Mbps, providing a substantial upgrade in wireless capabilities while NETGEAR WAC104 only supports Wi-Fi 5 with the speed 867Mbps at 5 Hz and 300 Mbps at 2.4 Hz. Besides, the EAP670 provides 204 m² coverage area while NETGEAR WAC 104 only provides 92 m² coverage area. Hence, we choose to use TP-Link EAP670. The EAP670's integration into Omada SDN facilitates centralized cloud management, enabling seamless control and monitoring of the network. With a comprehensive 5-year warranty, the EAP670 not only ensures a future-proof solution with Wi-Fi 6 but also provides reliability and support for an extended period, making it the ideal choice for a project requiring cutting-edge technology, advanced features, and robust performance.

Server:

For the servers used in this project, we have chosen [Cisco UCS C480M5 High-Performance Rack Server](#). The major difference between Cisco UCS C480M5 High-Performance Rack Server with other different brands we can compare is Lenovo ThinkSystem SR630.

Specification	Cisco UCS C480M5 High-Performance Rack Server	Lenovo ThinkSystem SR630
Form factor	4U rackmount	2U rackmount
Processors	Dual Intel Xeon Scalable processors (2nd generation)	Up to 2nd Gen Intel Xeon Scalable processors

		(up to 28 cores)
Memory	Up to 24 x 16GB DDR4 DIMMs for a total of 384GB	Up to 12 x 16GB DDR4 DIMMs for a total of 192GB
Storage	Up to 32x 2.5-inch hot-swappable SAS/SATA HDDs or SSDs	Up to 16x 2.5-inch hot-swappable SAS/SATA HDDs or SSDs
Networking	Dual 10GBASE-T Intel x550 network adapters	Dual 10GbE SFP+ ports
NVMe support	Up to 12 NVMe SSDs	Up to 6 NVMe SSDs
Management	Integrated management controller (iMC)	BMC with IPMItool

Based on the table above, the Cisco UCS C480M5 is a more powerful and scalable server than the Lenovo ThinkSystem SR630. It boasts a higher maximum CPU capacity, enabling it to handle more demanding workloads, and a higher maximum memory capacity, which can help to improve performance for memory-intensive applications. Additionally, it offers more storage options, providing the flexibility to store more data and choose the most suitable storage type. Furthermore, it features more networking and expansion slots, allowing for future expansion to meet evolving needs. However, it comes at a higher cost compared to its counterpart.

Firewall:

For firewall hardware, an additional useful security device, we select [FS NSG-5100 Next-Generation Firewall](#) in our project by comparing it with other different brand which is [NETGEAR FVS318N-100](#).

Specification	FS NSG-5100 Next-Generation Firewall	NETGEAR FVS318N-100
Maximum Concurrent Sessions	2 million	0.006 million (6,000)
IPsec VPN (3DES) Throughput	3 Gbps	20 Mbps

Suitability for Large Encrypted Traffic	Yes, high throughput	Limited by lower throughput
Threat Detection and Prevention	More comprehensive	Limited

By looking at the major differences between these two brands of firewall, an additional useful security device, FS NSG-5100 Next-Generation Firewall is our best option by renouncing another brand of firewall which is NETGEAR FVS318N-100. FS NSG-5100 Next-Generation Firewall has higher maximum concurrent sessions. This makes FS NSG-5100 Next-Generation Firewall can handle a higher number of simultaneous connections or sessions at any given time. In terms of performance for FS NSG-5100 Next-Generation Firewall, it can produce a higher throughput. With high throughput value, FS NSG-5100 Next-Generation Firewall can handle a large volume of encrypted traffic without significant degradation in performance. Thus, the selection of FS NSG-5100 Next-Generation Firewall is suitable as it also provides more comprehensive threat detection and prevention and the cost for purchase is also appropriate by considering our budget as it is an extra device, we added in the faculty building.

Personal Computer (PC):

For the Personal Computer (PC) used in this project, we have chosen [HP EliteDesk 800 G9 Desktop](#). The major difference between HP EliteDesk 800 G9 Desktop with other different brands we can compare is Dell OptiPlex 5090 Desktop.

	HP EliteDesk 800 G9 Desktop	Dell OptiPlex 5090 Desktop
Processor	Up to 13th Gen Intel® Core™ i9 processors	Up to 11th Gen Intel® Core™ i9 processors
Memory	Up to 64GB DDR5 memory	Up to 64GB DDR4 memory
Security	HP Wolf Security for Business	Dell Data Security
Manageability	HP Intelligent Management Suite	Dell Client Management Console
Price	Starting at RM 3,699	Starting at RM 2,999

The comparison table highlights the clear advantages of the HP EliteDesk 800 G9 Desktop over the Dell OptiPlex 5090 Desktop. The HP EliteDesk 800 G9 Desktop boasts a superior processor that enhances responsiveness and processing speed, ample memory that enables seamless multitasking, and robust security features that safeguard against online threats. While the Dell OptiPlex 5090 Desktop offers a more affordable price, the superior performance and security of the HP EliteDesk 800 G9 Desktop makes it the more compelling choice for those seeking a powerful and reliable desktop PC.

Monitor:

For the Monitor used in this project, we have chosen HP V24i FHD Monitor. The major difference between HP V24i FHD Monitor with other different brands we can compare is Dell 24 Inch FHD Computer Monitor - P2422H.

Feature	HP V24i FHD Monitor	Dell 24 Inch FHD Computer Monitor - P2422H
Response time	5ms GtG	8 ms (grey-to-grey normal); 5 ms (grey-to-grey fast)
Input connector	1 VGA; 1 HDMI 1.4 (with HDCP support)	DisplayPort (DisplayPort 1.2 mode, HDCP 1.4), VGA, HDMI (HDCP 1.4)

Based on the comparison table, the main difference between the two monitors is the response time. The HP V24i has a 5ms response time, while the Dell P2422H has a 8ms response time. This means that the HP V24i will be able to display images more quickly and with less ghosting.

The Dell P2422H also has a wider range of input connectors, including DisplayPort, VGA, and HDMI. This means that it is compatible with a wider range of devices, such as laptops, gaming consoles, and older computers. However, the HP V24i is still a good choice, as it has all of the essential connectors and a faster response time.

Projector:

For the projectors used in this project, we have chosen **Epson VS250 SVGA**. The major difference between Epson VS250 SVGA with other different brands we can compare is BenQ HT2060.

Specification	BenQ HT2060	Epson VS250 SVGA
Brightness	2,300 Lumens (ANSI)	3,200 Lumens (ANSI)
Native Resolution	1080P (1920x1080)	SVGA (800x600)
Aspect Ratio	16:9	4:3 (SVGA)
Contrast Ratio	500,000:1	15,000:1
Light Source	4LED	210W UHE bulb
Lamp Life	Normal 20,000 hrs, Eco 30,000 hrs	6,000 hours / 10,000 hours (Eco)
Throw Ratio	1.15 ~ 1.5	1.45:1 (D:W)
Resolution Support	VGA (640 x 480) to 4K UHD (3840 x 2160)	720p, 1080i, 1080p/60, MAX 1400x1050
I/O Interface	HDMI x 2, USB Type A, RS232	HDMI, USB x 2, VGA In

Based on the table above, the Epson VS250 SVGA is more suitable due to its higher brightness level of 3,200 Lumens, which is beneficial in well-lit meeting rooms. Its SVGA resolution, while lower than the BenQ HT2060's 1080P, is sufficient for standard presentations and charts. The Epson VS250 is also lighter and more compact, making it easier to move between meeting rooms or for storage. Although the BenQ HT2060 offers higher resolution and contrast

ratio, these features are more critical in environments where detailed graphics and videos are frequently presented, such as in film or design industries. Therefore, for standard business presentations, the Epson VS250 SVGA's combination of brightness, portability, and adequate resolution makes it a more practical choice.

Smart

TV:

The [Samsung QN900B](#) stands out as the more suitable choice for a video conferencing room with its superior features and specifications when compared to the other brand of smart TV LG 75UR7550PSC. Firstly, the Samsung QN900B boasts an exceptionally high resolution of 7,680 x 4,320, providing a level of clarity and detail that surpasses the 4K resolution of the LG TV. This is particularly advantageous for video conferences where visual precision is crucial. The QLED display technology in the Samsung further enhances the viewing experience, offering vibrant colors and improved contrast compared to the traditional LCD technology in the LG model. Additionally, the Samsung TV excels in audio performance with a remarkable 90W sound output, ensuring that participants in the video conference can enjoy clear and immersive audio while the LG TV only provides 20W sound output. The Samsung QN900B also features HDMI (High Frame Rate) 4K 120Hz, a specification that contributes to future-proofing the setup for potential advancements in video conferencing equipment while the LG TV only provides 60Hz refresh rate.. Overall, the combination of higher resolution, superior display technology, and advanced audio capabilities make the Samsung QN900B the optimal choice for delivering a top-tier video conferencing experience in a room of 10m x 14m dimensions.

Printer:

In this project, we selected [Canon Color imageCLASS MF751Cdw](#) as the main model of printer placed in the office and IOT Lab. By comparing the Canon Color imageCLASS MF751Cdw and Epson EcoTank Pro ET-5850, one of the major differences we can see from these two printers is print memory size. Canon Color imageCLASS MF751Cdw does provide 1GB of memory but Epson EcoTank Pro ET-5850 does not provide any memory. This memory allows the printer to handle complex documents, images, or graphics without needing to reprocess data for each page and it will be very helpful for staff or lecturers to handle their work. Besides that, Canon Color imageCLASS MF751Cdw provides some features like cancel and favourite setting which cannot

be found in Epson EcoTank Pro ET-5850 printer. Since Canon Color imageCLASS MF751Cdw and Epson EcoTank Pro ET-5850 have provided similar features like auto 2-sided printing, black, white and color printing, so by looking the cost for both printers, the Canon Color imageCLASS MF751Cdw will be a most cost-effective solution on selecting the printer device compared to Epson EcoTank Pro ET-5850 printer as the price of the CanonColor imageCLASS MF751Cdw is RM1874 which is higher than RM3748 for Epson EcoTank Pro ET-5850 printer.

Connectors:

For the connectors used in this project, we have chosen **Cable Matters Cat 6 Pass Through RJ45 Connectors**. The major difference between Cable Matters Cat 6 Pass Through RJ45 Connectors with other different brands we can compare is Klein Tools VDV826-703 Pass-Thru Modular Data Plug.

Feature/Aspect	Cable Matters Cat 6 Pass Through RJ45 Connectors	Klein Tools VDV826-703 Pass-Thru Modular Data Plug
Compatibility	Compatible with Cat 6 and possibly lower categories	Compatible with multiple categories, including Cat 6
Ease of Crimping	Designed for easy and reliable crimping	Engineered for efficient, reliable crimping
Price Range	Generally affordable, varies by retailer	May be premium-priced, varies by retailer
Ease of Wiring Verification	Allows for easy verification before crimping	Clear design for easy wire order verification
Certifications/Standards	Meets standard industry certifications	Meets or exceeds industry standards

Special Features	Basic functionality for everyday use	May include additional features for professional use
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In comparing Cable Matters Cat 6 Pass Through RJ45 Connectors and Klein Tools VDV826-703 Pass-Thru Modular Data Plugs, both products offer high-quality build and are designed to accommodate Cat 6 and possibly lower category cables. Cable Matters connectors are recognized for their affordability and reliability, making them a suitable choice for both professional and everyday use. On the other hand, Klein Tools' connectors are noted for their exceptional durability and are often preferred in professional settings, potentially carrying a premium price tag. Both brands support stranded and solid conductors and feature a pass-through design that facilitates easy wiring verification and efficient crimping. While Cable Matters offers basic, reliable functionality, Klein Tools might include additional features catering to professional demands, adhering to or exceeding standard industry certifications.

Fiber Optic Cable:

For the projectors used in this project, we have chosen **SC-UPC APC Pigtail**. The major difference between SC-UPC APC Pigtail with other different brands we can compare is SC UPC Patch Cable.

Feature	SC UPC Patch Cable	SC-UPC APC Pigtail
Insertion Loss	0.15dB	Not more than 0.3dB
Return Loss	53.6dB	Not less than 50dB

Insertion Loss: The SC-UPC APC Pigtail has a lower insertion loss (0.15dB) compared to the SC UPC Patch Cable (not more than 0.3dB), indicating better signal transmission efficiency for the pigtail. **Return Loss:** The SC-UPC APC Pigtail has a higher return loss (53.6dB) compared to the SC UPC Patch Cable (not less than 50dB), indicating better signal reflection characteristics for the pigtail.

Meeting Minutes #1

Date/Time		13 NOV 2023 8.00pm	
Location		Online (Google Meet)	
Agenda		1. Understand the details about task 3	
		2. Suggest possible network devices to be included in project	
		3. Suggest the quantity for each device	
		4. Task distribution for all members	
Meeting Host		Koh Li Hui	
Attendance			
Name		Time	Reason For Absence
Yong Wern Jie		2000	-
Chen Pyng Haw		2000	-
Koh Li Hui		2000	-
Tan Jun Yuan		2000	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discussion for task 3	- Li Hui shared the information about task 2 and its rubric.	Li Hui
		- All members read and investigate the task carefully.	
		- Solved the problem raised by group members together through discussion.	
2	Suggestion for network devices that will be	- Jun Yuan and Wern Jie suggest the network devices	

	included in project	that will be used in ground floor	All members
		- Pyng Haw and Li Hui suggest the network devices that will be used in first floor	
3	Suggest the quantity for each devices	- Wern Jie and Jun Yuan suggest the quantity for each required network devices in ground floor	All members
		- Pyng Haw and Li Hui suggest the quantity for each required network devices in ground floor	
4	Task distribution	- Jun Yuan distributed the task to all members equally. - All members had to do research to select the model and brand of network devices that best suited to the project. - Wern Jie was assigned to do research on the brand/model of router, server and personal computer. - Li Hui was assigned to do research on the brand/model of switch, firewall and printer.	All members

		- Pyng Haw was assigned to do research on the brand/model of patch panel, modem, and projector. - Jun Yuan was assigned to do research on the brand/model of cable, wireless access point and smart TV.	
5	Meeting Ended	- At 2230, the meeting ended after all discussions were done.	All members

Meeting Minutes #2

Date/Time		22 NOV 2023 9.00pm	
Location		Online (Google Meet)	
Agenda		1. Review the current progress for each group member.	
		2. Determine the most suitable devices used in our floorplan.	
		3. Finalize the total prices for every device needed.	
Meeting MC		Koh Li Hui	
Attendance			
Name		Time	Reason For Absence
Yong Wern Jie		2100	-
Chen Pyng Haw		2100	-
Koh Li Hui		2100	-
Tan Jun Yuan		2100	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Review the current progress for each group member	- Wern Jie shared his part.	All members
		- Pyng Haw shared his part.	
		- Jun Yuan shared his part.	
		- Li Hui shared her part.	
2	Determine the most suitable network devices	- Li Hui wonders whether the selected devices can work seamlessly with other network devices.	All members
		- Pyng Haw determines whether the chosen brand of devices	

		can accommodate future growth in terms of the number of users, devices, and network traffic.	
		- Wern Jie reminds the other group members of the types of network devices and features needed based on the network design.	
		- Jun Yuan suggests adding additional devices since there is enough budget.	
		- All members shared their opinions by comparing different brands of network devices and considering several factors.	
3	Finalize the total prices for every device needed	- Jun Yuan and Pyng Haw calculate the prices corresponding to each device.	Wern Jie
		- Li Hui checks the total prices are within the budget given for our network plan.	
		- Wern Jie facilitates a discussion to finalize the decision on the most suitable network devices.	
4	Meeting ended	- At 2315, the meeting ended after all discussions were done.	All members

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